

16-Week Delivery Schedule & Pacing Guide

Session-by-session breakdown for weekday, weekend, and online cohorts

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1 How to Read This Schedule

This document provides a **week-by-week breakdown** of the 16-week curriculum. Each section includes:

- **Content:** Notebooks and exercises to cover
- **Learning Objectives:** What students must master by week’s end
- **Estimated Hours:** Total instructional + study time
- **Cohort Notes:** How each cohort type (weekday, weekend, online) adapts the timing

1.1 Cohort Types

Cohort	Sessions/Week	Hours/Session	Total/Week	Best For
Weekday	3 (Mon, Wed, Fri)	1.5 hours	~4.5 hours	Working professionals, flexible schedules
Weekend	1 (Saturday)	4 hours	~4 hours	9-to-5 workers; includes optional Sunday office hours (1 hr)
Online	Async + 1 live	Videos + 2hr Q&A	~5 hours	Distributed learners, different time zones

2 PHASE 1: FOUNDATIONS (Weeks 1-4)

2.1 Week 1: Welcome, Setup & Environment

Goal: Students have working Python development environment; understand why Python for AI

Content:

- Welcome Guide (PDF) (1 hour)
- Setup Guide (PDF) (2 hours)
- **Live Activity:** Install uv, Python 3.12, VS Code; create first virtual environment

Learning Objectives:

- Understand the Python ecosystem for AI (PyTorch, TensorFlow, Hugging Face)
- Install and manage Python versions with `uv`
- Create isolated environments using `pyproject.toml`
- Open and navigate a Jupyter notebook

Estimated Hours: 3 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon (Welcome) + 1.5hr Wed (Setup) = 3 hrs
- **Weekend:** 3hr Saturday morning session
- **Online:** Pre-recorded videos (watch 1 hr) + 2hr live Q&A Saturday morning

Homework:

- ☐ Successfully run `uv sync` in a test project
- ☐ Create a GitHub account; fork the course repo
- ☐ Verify Jupyter launches and can run `print("Hello, Ghana!")`

2.2 Week 2: Professional Version Control with Git & GitHub

Goal: Students use Git/GitHub as AI engineers (not just web developers)

Content:

- Git `02-version-control/Git_Github.qmd` GitHub Guide (PDF) (3 hours)
- **Live Activity:** Clone a repo, make a branch, commit, push, create a pull request

Learning Objectives:

- Understand Git vs GitHub distinction
- Create meaningful commit messages
- Use branches for experiment tracking
- Build a GitHub portfolio for AI employers
- Understand merge conflicts and resolution

Estimated Hours: 3 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed
- **Weekend:** 3hr Saturday (or split across Sat + Sun if needed)
- **Online:** 2hr live session + 1hr async homework

Homework:

- ☐ Create 3 feature branches and merge to main
- ☐ Write 5 meaningful commits (not “fix typo”)
- ☐ Review a peer’s PR (if working in cohort)

2.3 Week 3: Python Basics - Variables & Data Structures

Goal: Smooth transition from JavaScript to Pythonic syntax

Content:

- Python Basics Part 1 (PDF) (4 hours)
- **Live Activity:** Code-along; walk through 4 exercises (lists, tuples, dicts, sets)

Learning Objectives:

- Master Python naming conventions (snake_case)
- Create and manipulate lists, dicts, sets, tuples
- Index and slice sequences correctly
- Understand mutable vs immutable types
- Use f-strings for string formatting

Estimated Hours: 4 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon (Variables) + 1.5hr Wed (Data Structures) + 1hr Fri (exercises)
- **Weekend:** 2hr Saturday + 2hr Sunday
- **Online:** 3hr async videos + 1hr live Q&A Tuesday evening

Homework: - [] Complete all 4 exercises in the notebook with correct solutions - [] Write 3 additional list comprehensions (no loops) - [] Quiz (5 questions on indexing and slicing)

2.4 Week 4: Functions, Scope & Pythonic Patterns (+ Mini Demo Day 1)

Goal: Master functions and Pythonic idioms; deliver first mini-project

Content:

- Python Basics Part 2 (PDF) (5 hours)
- Python Basics Part 3 (PDF) (5 hours)
- Python Exercises (PDF) (2 hours - supplementary)
- **Live Activity:** Code-along; troubleshoot scope issues; present Mini Demo Day 1 projects

Learning Objectives:

- Write clean, reusable functions with docstrings
- Understand Python scope (LEGB rule)
- Use *args, **kwargs, and default arguments
- Write list/dict/set comprehensions fluently
- Understand OOP basics: classes, inheritance, methods
- Handle exceptions with try/except

Estimated Hours: 12 hours total (4.5 hours live + 7.5 hours study + exercises)

Cohort Notes:

- **Weekday:** 1.5hr Mon (Functions) + 1.5hr Wed (Comprehensions) + 1.5hr Fri (OOP) + 1.5hr Fri (Demo Day prep)
- **Weekend:** 3hr Saturday (Functions + Comprehensions) + 3hr Sunday (OOP + Demo Day prep)
- **Online:** 5hr async videos + 2hr live Wed evening (Q&A) + async demo prep

Mini Demo Day 1: Simple Logic Script

- **Timing:** Friday late afternoon or Saturday for weekend cohort
- **Format:** Students present 10-min demo of working Python CLI tool
- **Examples:** Voting system, expense tracker, MTN subscriber manager, student grade calculator
- **Submission:** Working Python script (.py file) + README.md + GitHub repo link
- **Grading:** See Milestone Rubrics (PDF) for scoring rubric

Homework:

- ☐ Complete exercises 1–6 from Python Exercises (PDF)
- ☐ Build your Mini Demo Day 1 project
- ☐ Practice 1-min elevator pitch for your project

3 PHASE 2: ANALYTICS (Weeks 5-8)

3.1 Week 5: Pandas Fundamentals - DataFrames & Series

Goal: Load, explore, and manipulate data with Pandas

Content:

- Pandas Part 1 (PDF) (5 hours)
- **Live Activity:** Load CSV, explore with `.head()`, `.describe()`, `.info()`, practice indexing

Learning Objectives:

- Load data from CSV, JSON, Excel
- Understand DataFrame vs Series
- Use `.loc[]` (label-based) and `.iloc[]` (position-based) indexing
- Select columns and rows using boolean masks
- Handle missing values and duplicates
- Understand data types and convert when needed

Estimated Hours: 5 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1hr Fri
- **Weekend:** 2hr Saturday + 2hr Sunday

- **Online:** 3hr async videos + 1.5hr live Wed Q&A

Homework:

- ☐ Load 3 different CSV files; describe their shape and dtypes
- ☐ Use `.loc[]` and `.iloc[]` to answer 5 data questions
- ☐ Fix a messy dataset (convert dtypes, handle NaN)

3.2 Week 6: Data Visualization & Transformation

Goal: Create publication-ready visualizations; understand data storytelling

Content:

- Pandas Part 2 (PDF) (5 hours)
- **Live Activity:** Create line plots, scatter plots, histograms, heatmaps

Learning Objectives:

- Use Matplotlib for low-level control
- Use Seaborn for statistical beauty
- Create multi-panel figures
- Visualize distributions, correlations, time series
- Understand when to use which plot type
- Add titles, labels, legends; make plots publication-ready

Estimated Hours: 5 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1hr Fri
- **Weekend:** 2hr Saturday + 2hr Sunday
- **Online:** 3hr async videos + 1.5hr live Wed Q&A

Homework:

- ☐ Recreate 5 plots from the notebook
- ☐ Create 3 new visualizations on Ghana tourism or health data
- ☐ Quiz: “What plot would you use to show correlation between 50 variables?”

3.3 Week 7: Tidy Data, Groupby & Merging

Goal: Reshape and combine datasets; prepare data for machine learning

Content:

- Pandas Part 3 (PDF) (5 hours)
- Pandas Exercises (PDF) (2 hours - supplementary)
- **Live Activity:** Melt, pivot, groupby, merge operations

Learning Objectives:

- Understand tidy data principles (Wickham 2014)
- Use `.melt()` to reshape wide → long format
- Use `.pivot_table()` to reshape long → wide format
- Apply groupby + aggregation patterns
- Merge DataFrames using `.merge()`, `.join()`, `.concat()`
- Identify and fix multi-table databases

Estimated Hours: 7 hours

Cohort Notes: - **Weekday:** 1.5hr Mon + 1.5hr Wed + 1.5hr Fri + 1hr supplementary -

Weekend: 3hr Saturday + 2hr Sunday - **Online:** 4hr async videos + 2hr live Fri evening (Q&A)

Homework: - [] Melt 2 wide datasets to long format - [] Groupby + aggregate to answer 5 business questions (Ghana housing, health, telecom) - [] Merge 2 datasets; verify no data loss or duplication

3.4 Week 8: Mini Demo Day 2 - EDA Report (+ Visualization Review)

Goal: Deliver first analysis project; refine visualization and communication skills

Content:

- Data Analysis Presentation Slides (PDF) (1 hour - recap)
- **Live Activity:** Group work on Mini Demo Day 2 projects; peer review

Learning Objectives:

- Complete end-to-end EDA on a real dataset
- Write clear data narratives
- Create publication-ready figures
- Present findings to non-technical audience

Estimated Hours: 6 hours (live + project time)

Cohort Notes:

- **Weekday:** 1.5hr Mon (EDA tips) + ongoing project work + 1.5hr Fri (presentations)
- **Weekend:** 2hr Saturday (tips) + async project work + 2hr Sunday (presentations)
- **Online:** 1hr async tips video + async project + 2hr live Saturday evening (presentations)

Mini Demo Day 2: Exploratory Data Analysis Report

- **Timing:** End of week 8 (Friday afternoon or Sunday)
- **Format:** Jupyter notebook + PDF report + 5-slide summary
- **Dataset:** Ghana health, education, telecom, or housing data (provided or student-chosen)
- **Deliverables:**
 1. Data exploration (10 questions answered)
 2. 5+ visualization with interpretations
 3. Key findings summary (1-page executive summary)
- **Submission:** GitHub repo + PDF report
- **Grading:** See Milestone Rubrics (PDF)

Homework:

- ☐ Complete exercises 1–6 from Pandas Exercises (PDF)
- ☐ Build your Mini Demo Day 2 EDA project
- ☐ Practice 5-min elevator pitch

4 PHASE 3: INTELLIGENCE (Weeks 9-12)

4.1 Week 9: Machine Learning Fundamentals & Linear Regression

Goal: Understand ML workflow; implement first predictive model

Content:

- ML Part 1 (PDF) (5 hours)
- **Live Activity:** Code-along train/test split, fit linear regression, interpret coefficients

Learning Objectives:

- Understand ML problem framing (supervised vs unsupervised)
- Implement proper train/test split (no data leakage)
- Fit linear regression models
- Calculate loss functions (MSE, RMSE)
- Interpret model coefficients
- Understand underfitting vs overfitting
- Create learning curves

Estimated Hours: 5 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1hr Fri
- **Weekend:** 2.5hr Saturday + 1.5hr Sunday
- **Online:** 3hr async + 1.5hr live Wed Q&A

Homework:

- ☐ Fit linear regression on 3 different datasets
- ☐ Calculate residuals and interpret them
- ☐ Quiz: "Why do we need train/test split?"

4.2 Week 10: Classification & Model Evaluation

Goal: Build classifiers; master evaluation metrics

Content:

- ML Part 2 (PDF) (5 hours)
- **Live Activity:** Train logistic regression and decision trees; build confusion matrices

Learning Objectives:

- Understand classification vs regression
- Train logistic regression and decision trees
- Interpret decision boundaries
- Create confusion matrices
- Calculate precision, recall, F1, ROC-AUC
- Choose metrics based on use case (imbalanced data, cost asymmetry)
- Handle overfitting with tree depth control

Estimated Hours: 5 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1hr Fri
- **Weekend:** 2.5hr Saturday + 1.5hr Sunday
- **Online:** 3hr async + 1.5hr live Wed Q&A

Homework:

- ☐ Train 3 classifiers on iris/titanic datasets
- ☐ Build confusion matrices and calculate 5 metrics
- ☐ Quiz: "When would you optimize for recall instead of precision?"

4.3 Week 11: Ensemble Methods & Hyperparameter Tuning

Goal: Master advanced ML techniques; improve model performance

Content:

- ML Part 3 (PDF) (5 hours)
- ML Exercises (PDF) (2 hours - supplementary)
- **Live Activity:** Train Random Forest and XGBoost; use GridSearchCV for tuning

Learning Objectives:

- Understand ensemble methods: bagging vs boosting
- Implement Random Forest and XGBoost
- Extract and interpret feature importance
- Use GridSearchCV for hyperparameter tuning
- Avoid overfitting with regularization and cross-validation
- Compare model performance using multiple metrics

Estimated Hours: 7 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1.5hr Fri + 1hr supplementary
- **Weekend:** 3hr Saturday + 2hr Sunday
- **Online:** 4hr async + 2hr live Fri evening (Q&A)

Homework:

- ☐ Complete exercises 1-5 from ML Exercises (PDF)



Tune Random Forest on Ghana housing data



Compare XGBoost vs Random Forest on evaluation metrics

4.4 Week 12: Mini Demo Day 3 - Predictive Model Build

Goal: Deliver end-to-end ML project; prepare for capstone work

Content:

- ML Presentation Slides (PDF) (1 hour - recap)
- **Live Activity:** Model selection workshop; peer code review

Learning Objectives:

- Complete end-to-end ML pipeline
- Make defensible model choices
- Interpret predictions and explain limitations
- Document code and decisions

Estimated Hours: 8 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon (Model selection) + ongoing project + 1.5hr Fri (presentations)
- **Weekend:** 2hr Saturday (tips) + async project + 2hr Sunday (presentations)
- **Online:** 1hr async video + async project + 2hr live Saturday evening

Mini Demo Day 3: Predictive Model Build

- **Timing:** End of week 12
- **Format:** Trained model (.pkl) + evaluation report + code notebook
- **Dataset:** Ghana housing prices, student performance, or telecom churn (provided)
- **Deliverables:**
 1. Preprocessed dataset with feature engineering
 2. 3+ trained models (logistic regression, decision tree, Random Forest)
 3. Evaluation report: metrics, confusion matrix, ROC curve
 4. Model interpretation: feature importance, SHAP values (if time)
- **Submission:** GitHub repo + saved model + evaluation notebook
- **Grading:** See Milestone Rubrics (PDF)

Homework:



Complete supplementary ML exercises if needed

- ☐ Build your Mini Demo Day 3 model
- ☐ Prepare 5-min presentation

5 PHASE 4: ADVANCED AI (Weeks 13-16)

5.1 Week 13: Deep Learning Fundamentals & Neural Networks

Goal: Understand neural networks; implement training loops in PyTorch

Content:

- Deep Learning Part 1 (PDF) (6 hours)
- **Live Activity:** Code-along simple neural networks; debug training loops on Colab

Learning Objectives:

- Understand neurons, layers, weights, biases
- Implement forward pass
- Understand backpropagation conceptually
- Use PyTorch to build and train networks
- Choose activation functions (ReLU, Sigmoid, Softmax)
- Understand loss functions and optimizers (SGD, Adam)
- Debug overfitting with regularization and dropout

Estimated Hours: 6 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1.5hr Fri (+ optional Colab homework)
- **Weekend:** 3hr Saturday + 2hr Sunday (+ Colab homework)
- **Online:** 4hr async videos + 1.5hr live Wed Q&A (+ Colab homework)

Note on GPU: Deep learning requires GPU access. Students should:

- Use Google Colab (free) for hands-on exercises
- Download trained checkpoints to run locally if needed

Homework:

- ☐ Run 3 training loops on MNIST in Google Colab
- ☐ Experiment with learning rates: `[0.0001, 0.001, 0.01, 0.1]`
- ☐ Plot learning curves; identify underfitting/overfitting

5.2 Week 14: CNNs, Image Classification & Transfer Learning

Goal: Build image classifiers; understand transfer learning

Content:

- Deep Learning Part 2 (PDF) (6 hours)
- **Live Activity:** Code-along CNN architecture; train on traffic signs; visualize learned features

Learning Objectives:

- Understand convolutions and pooling
- Build CNNs for image classification
- Use data augmentation to improve generalization
- Apply transfer learning: load pre-trained weights, fine-tune
- Visualize learned filters
- Handle imbalanced datasets

Estimated Hours: 6 hours

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1.5hr Fri (+ Colab homework)
- **Weekend:** 3hr Saturday + 2hr Sunday (+ Colab homework)
- **Online:** 4hr async videos + 1.5hr live Wed Q&A (+ Colab homework)

Homework:

- ☐ Train a CNN on GTSRB (German traffic signs) in Colab
- ☐ Fine-tune a pre-trained ResNet on cassava disease data
- ☐ Visualize 5 learned filters from the first convolution layer

5.3 Week 15: Agentic AI, LLMs & Orchestration

Goal: Understand the future of AI; build simple agents

Content:

- Deep Learning Part 3 (PDF) (6 hours - transfer learning + deployment)
- Agentic AI Presentation Slides (PDF) (1 hour - slide recap)
- Intro to Agentic AI (PDF) (2 hours)
- Intro to CodeTrain (PDF) (2 hours)

- OpenCode Student Guide (PDF) (2 hours)
- CodeTrain Tests Guide (PDF) (2 hours)
- **Live Activity:** Build a simple chatbot or research agent

Learning Objectives:

- Understand LLMs and their capabilities/limitations
- Understand Retrieval Augmented Generation (RAG)
- Understand agentic workflows: decision loops, tool use
- Build a simple agent using CodeTrain or similar framework
- Understand model orchestration and scaling

Estimated Hours: 15 hours (mix of async videos + live coding)

Cohort Notes:

- **Weekday:** 1.5hr Mon + 1.5hr Wed + 1.5hr Fri (Week 15) + 1.5hr Mon + 1.5hr Wed (Week 16 project)
- **Weekend:** 3hr Saturday + 3hr Sunday (split across weeks 15-16)
- **Online:** 5hr async videos (week 15) + 2hr live Wed Q&A + async project (week 16)

Homework:

- ☐ Watch OpenCode/CodeTrain tutorials
- ☐ Build a simple chatbot using LLM API
- ☐ Deploy the agent and test with 5 sample inputs

5.4 Week 16: Final Demo Day - Capstone AI Solution

Goal: Deliver portfolio-ready AI project; celebrate learning

Content:

- **Live Activity:** Team capstone presentations; peer feedback; instructor Q&A

Learning Objectives:

- Complete end-to-end AI solution (image classifier, NLP agent, or predictive model)
- Present technical work to non-technical audience
- Document code and decisions for future reference
- Reflect on learning and next steps

Estimated Hours: 8 hours (presentations + preparation)

Capstone Project: Final Demo Day

- **Teams:** 2–3 students per team
- **Project Duration:** Weeks 13–16 (4 weeks to build)
- **Deliverables:**
 1. GitHub repository with documented code
 2. Trained model (or deployed agent) + inference script
 3. 10-slide presentation deck
 4. 3-page technical report
- **Presentation:** 15 min demo + 10 min Q&A
- **Grading:** See Milestone Rubrics (PDF)
- **Portfolio Goal:** Students should be able to show this project to employers

Capstone Project Ideas:

- **Image Classification:** Classify crops (healthy vs diseased) for Ghanaian farmers
- **NLP Agent:** Customer service chatbot for MTN or mPharma
- **Predictive Model:** Predict student dropout risk or loan default
- **Time Series:** Forecast electricity demand for Accra
- **Recommendation System:** Recommend educational content to students

6 Cohort-Specific Pacing Notes

6.1 Weekday Cohort (Mon/Wed/Fri afternoons)

- **Advantages:** Consistent rhythm; students can ask questions immediately
- **Challenges:** Students juggle work/study; limited async flexibility
- **Recommendation:** 1.5-hour sessions max; record all sessions for async review
- **Office Hours:** Friday 4–5pm or Monday morning (optional)

6.2 Weekend Cohort (Saturday mornings)

- **Advantages:** Immersive; dedicated time; can go deeper
- **Challenges:** Single session may drag; students tired after 4 hours
- **Recommendation:** 4-hour session with 15-min break midway; optional 1-hour Sunday review
- **Office Hours:** Sunday afternoons (optional, 1 hour)

6.3 Online Cohort (Async + live Q&A)

- **Advantages:** Flexible; students watch videos at own pace; distributed access
- **Challenges:** Risk of falling behind; less interaction; needs structure

- **Recommendation:**

- Pre-record all lectures (can use PowerPoint + voiceover or Jupyter + Loom)
- Require weekly video submissions (students screen-record exercises)
- Live Q&A session mid-week (same 5-minute questions answered live)
- Assign peer reviewers for exercise solutions

- **Office Hours:** Async Slack channel + 1-hour Tuesday evening live (optional)

7 Assessment Timeline

Week	Gate / Milestone
4	Mini Demo Day 1: Python Logic Script
8	Mini Demo Day 2: Exploratory Data Analysis
12	Mini Demo Day 3: Predictive Model Build
16	Final Demo Day: Capstone AI Solution

Grading Rubrics: See Milestone Rubrics (PDF) for detailed scoring criteria.

8 Catch-Up & Remediation

8.1 If a student falls behind:

1. **Week 1-4:** Assign extra Python exercises. Do NOT proceed to Pandas if they're still confused on variables/scope.
2. **Week 5-8:** Offer supplementary Pandas exercises. Pair with a study buddy.
3. **Week 9-12:** Assign ML exercises; offer 1:1 tutoring on model evaluation.
4. **Week 13-16:** Simplify capstone scope (use pre-trained model, focus on deployment/presentation).

8.2 If a student is ahead:

1. Encourage them to help peers with exercises
2. Suggest extended projects: "Build this on a different dataset" or "Deploy this model as an API"
3. Introduce research papers related to the week's topic

9 Resources & Tools

For Students:

- Python IDE: VS Code or JupyterLab
- GPU: Google Colab (free GPU for deep learning)
- Version Control: GitHub
- Chat/Q&A: Slack or Discord channel

For Instructors:

- Curriculum Strategy: Curriculum Strategy (PDF)
- Assessment Rubrics: Milestone Rubrics (PDF)
- Trainer Guide: Trainer Guide (PDF) (common struggles, interventions)
- Presentation Slides: Available in each module folder (Presentation Slides (PDF))

This schedule is designed to be flexible. Adjust week-by-week based on cohort pace and learning gates. The goal is mastery, not rushing.